

AQbD Analytical Method Development Report

Analytical Quality by Design (AQbD) / Design of Experiments (DOE) Studio V1

Project Code:	Paracetamol_Dev	Date / Time:	2026-10-02 02:08:28
Project Name:	HPLC Paracetamol Method Development	Software:	AQbD Studio V1 (ICH-Aligned)

1. Executive Summary

This report documents the systematic, science- and risk-based development of the analytical procedure following ICH Q14 guidelines. The workflow progresses from defining the Analytical Target Profile (ATP) and initial Failure Mode and Effects Analysis (FMEA) risk assessment, to multivariate Design of Experiments (DOE), rigorous ANOVA and diagnostic model verification, multi-response desirability optimization, design space boundaries determination, and experimental confirmation.

2. Analytical Target Profile (ATP)

Parameter	Criterion	Unit	Acceptance Limits / Target
Retention Time	RANGE	min	[4.0 to 6.0]
Assay	TARGET	%	= 100.0
Tailing Factor	MAXIMUM	-	<= 2.0
Theoretical Plates	MINIMUM	-	>= 2000.0

3. Risk Assessment (FMEA)

Parameter	Severity (S)	Occurrence (O)	Detectability (D)	RPN	Priority
Mobile Phase Composition	8	6	5	240	Critical

4. Experimental Factors (DOE Parameters)

Code	Factor Name	Unit	Low (-1)	High (+1)	Role
A	Mobile Phase Composition	%	60.0	70.0	CRITICAL
B	Flow Rate	mL/min	0.8	1.2	CRITICAL
C	Column Temperature	°C	25.0	35.0	CRITICAL

5. Experimental Responses & Goals

Code	Response Name	Unit	Goal	Limits / Target	Weight
Y1	Resolution	-	MAXIMIZE	-	5
Y2	Tailing Factor	-	MINIMIZE	-	4
Y3	Retention Time	min	TARGET	= 5.0	4
Y4	Theoretical Plates	plates	MAXIMIZE	-	3

6. Design of Experiments (DOE) Execution

Active Design (ID 13): **FULL_FACTORIAL** | Runs: **11** | Random Seed: **43**

7. Statistical Modeling & Optimization

Active Response Models:

- ID #5 - Response ID 11 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #10 - Response ID 11 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #11 - Response ID 11 (INTERACTIONS): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #12 - Response ID 11 (QUADRATIC): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #13 - Response ID 12 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #14 - Response ID 12 (INTERACTIONS): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #15 - Response ID 12 (QUADRATIC): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #16 - Response ID 13 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #17 - Response ID 13 (INTERACTIONS): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #18 - Response ID 13 (QUADRATIC): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #19 - Response ID 14 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #20 - Response ID 14 (INTERACTIONS): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #21 - Response ID 14 (QUADRATIC): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #24 - Response ID 11 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #25 - Response ID 11 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #26 - Response ID 12 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #27 - Response ID 13 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$
- ID #28 - Response ID 14 (FIRST_ORDER): $R^2 = \text{N/A}$ | Adj $R^2 = \text{N/A}$ | Pred $R^2 = \text{N/A}$

Active Multi-Response Optimal Solution (Run #25):

Overall Desirability (D): **0.6253**

Optimal Setpoint: A = 66.271, B = 0.800, C = 35.000

8. Design Space & Proven Acceptable Range (PAR)

Full 3D Design Space Analysis (Run #34):

- Grid Vertices Evaluated: 8000
- Feasible/Acceptable Region: 6234 points (77.9%)
- Unacceptable Region: 1766 points
- Extrapolation Allowed: Zero

Projected Feasible Spans (Envelope):

- A = 60.00 – 70.00
- B = 0.80 – 1.20
- C = 25.00 – 35.00

Note: The projected spans represent the outer envelope of feasibility across the evaluated dimensions. They do not constitute a fully uncoupled Cartesian operating box; some combinations near the edges may still fail constraints depending on interactions. (Note: 2D cross-sectional slices visualized in the software are distinct from this full 3D volume analysis).

9. Confirmation Validation

Confirmation Run ID: CR-3 — ■ Software Simulation Check

- Y1: Predicted = 2.599 | Simulated = 2.599 | Diff = 0.000
- Y2: Predicted = 1.101 | Simulated = 1.101 | Diff = 0.000
- Y3: Predicted = 5.324 | Simulated = 5.324 | Diff = 0.000
- Y4: Predicted = 2698.378 | Simulated = 2698.378 | Diff = 0.000

Verdict: Software Verification Passed (Data Source: SIMULATED)

Confirmation Run ID: CR-4 — ■ Software Simulation Check

- Y1: Predicted = 2.599 | Simulated = 2.599 | Diff = 0.000
 - Y2: Predicted = 1.101 | Simulated = 1.101 | Diff = 0.000
 - Y3: Predicted = 5.324 | Simulated = 5.324 | Diff = 0.000
 - Y4: Predicted = 2698.378 | Simulated = 2698.378 | Diff = 0.000
- Verdict: Software Verification Passed (Data Source: SIMULATED)*

Confirmation Run ID: CR-5 — ■ Software Simulation Check

- Y1: Predicted = 2.599 | Simulated = 2.599 | Diff = 0.000
 - Y2: Predicted = 1.101 | Simulated = 1.101 | Diff = 0.000
 - Y3: Predicted = 5.324 | Simulated = 5.324 | Diff = 0.000
 - Y4: Predicted = 2698.378 | Simulated = 2698.378 | Diff = 0.000
- Verdict: Software Verification Passed (Data Source: SIMULATED)*

Confirmation Run ID: CR-6 — ■ Software Simulation Check

- Y1: Predicted = 2.599 | Simulated = 2.599 | Diff = 0.000
 - Y2: Predicted = 1.101 | Simulated = 1.101 | Diff = 0.000
 - Y3: Predicted = 5.324 | Simulated = 5.324 | Diff = 0.000
 - Y4: Predicted = 2698.378 | Simulated = 2698.378 | Diff = 0.000
- Verdict: Software Verification Passed (Data Source: SIMULATED)*

10. Conclusion

The analytical method was developed and evaluated using systematic AQbD principles. The workflow produced model-based predictions, multi-response optimization results, and a computed multidimensional design space. Simulated confirmation records verify the software calculation workflow only and do not constitute empirical laboratory confirmation.

Generated automatically by AQbD Studio V1. Aligned with ICH Q8/Q9/Q14 Enhanced Lifecycle Guidelines.